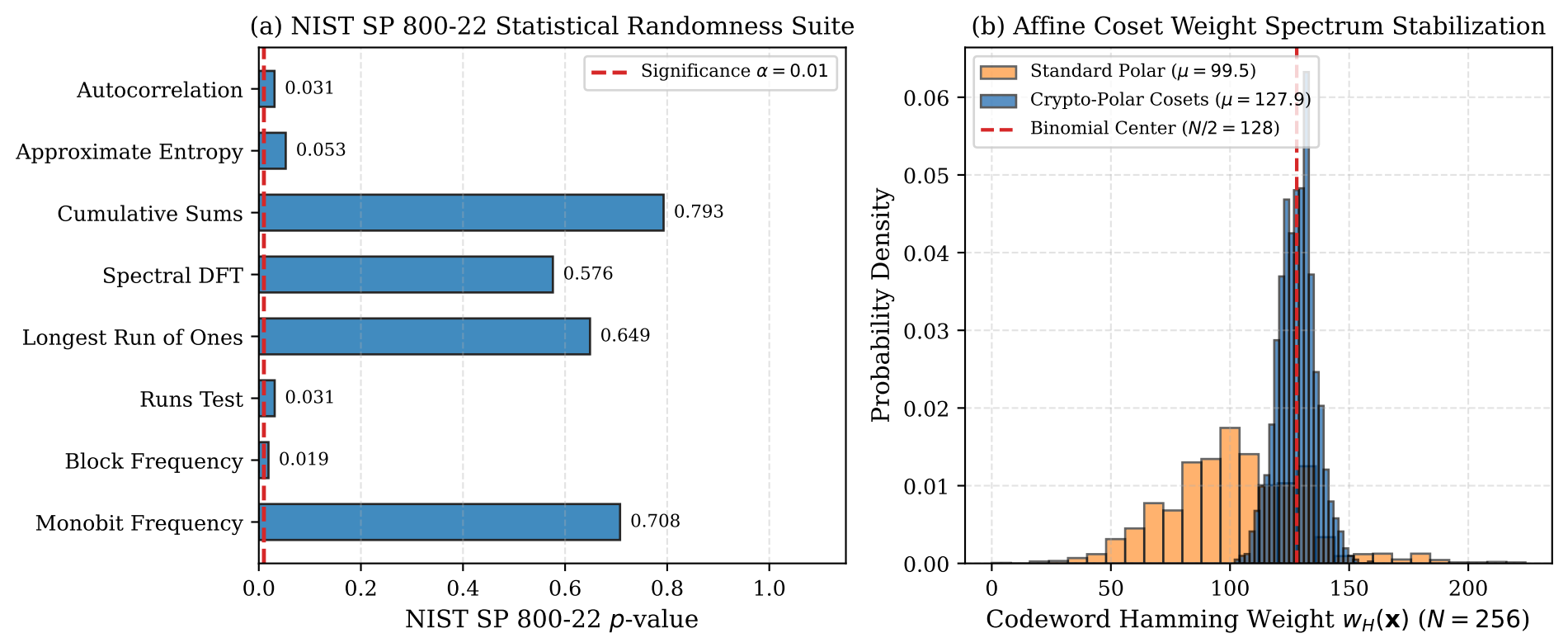
(a) NIST SP 800-22 Statistical Randomness Suite
Significance $\alpha = 0.01$
Autocorrelation
Approximate Entropy
Cumulative Sums
Spectral DFT
Longest Run of Ones
Runs Test
Block Frequency
Monobit Frequency
0.031
0.053
0.793
0.576
0.649
0.031
0.019
0.708
0.0
0.2
0.4
0.6
0.8
1.0
NIST SP 800-22 p-value
(b) Affine Coset Weight Spectrum Stabilization
Standard Polar ($\mu = 99.5$)
Crypto-Polar Cosets ($\mu = 127.9$)
Binomial Center ($N/2 = 128$)
Probability Density
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0
50
100
150
200
Codeword Hamming Weight $w_H(\mathbf{x})$ ($N = 256$)